Technical Volume

Aerivio Systems — SBIR Phase I Proposal

Topic N251-042: Autonomous Acoustic Threat Classification at the Tactical Edge

Period of Performance: 6 months · Requested Funding: $150,000 · Distribution: Proprietary

Executive Summary

Unmanned surface vessels (USVs) increasingly operate beyond reliable communications, yet current acoustic-threat detection depends on shore-side processing that introduces prohibitive latency. Aerivio Systems proposes a quantized, edge-deployable acoustic classifier that identifies and prioritizes sub-surface and surface threats in **under 100 milliseconds**, entirely on-vessel. This is a paradigm shift from centralized triage to autonomous tactical response.

Our Phase I effort will demonstrate feasibility of a 4-bit quantized convolutional network achieving *greater than 92%* classification accuracy on the DARPA-released hydrophone corpus while fitting within the 2-watt power envelope of a fielded USV compute module. Success in Phase I directly enables a Phase II at-sea demonstration and a clear transition path to Navy programs of record.

The remainder of this volume establishes the operational problem, details the technical approach and its three innovations, presents preliminary evidence of feasibility, and lays out a task-by-task work plan, a quantitative evaluation methodology, and a risk-mitigation strategy for the six-month period of performance.

1. Problem and Significance

The Navy fields a rapidly growing fleet of autonomous surface platforms, but their acoustic sensing pipelines remain tethered to intermittent SATCOM links. When a link degrades — the norm in contested or high-sea-state environments — classification stalls and the vessel loses the ability to distinguish a biologic from a torpedo. The operational consequence is severe: autonomy is throttled precisely when it matters most.

Consider a representative scenario. A USV on a persistent picket mission detects an ambiguous acoustic signature. Under today's architecture, the raw stream is buffered and, if bandwidth permits, relayed to a shore station for classification; the round trip can exceed several seconds. A torpedo closing at 40 knots covers more than 60 meters in that window. The vessel needs a classification decision at the tactical edge, in tens of milliseconds, with no assumption of connectivity.

The significance extends beyond a single platform. Every autonomous hull the Navy fields inherits this same dependency. An edge-native classifier that fits the existing 2-watt compute budget would retrofit the deployed fleet without new hardware, converting a fleet-wide vulnerability into a fleet-wide capability.

Closing this gap is fundamentally a machine-learning-systems problem. State-of-the-art acoustic classifiers are accurate but far too large and power-hungry for a fielded module. Naïve compression collapses accuracy on exactly the confusable classes that matter. Aerivio's innovation is a co-designed quantization-and-pruning pipeline that preserves discriminative fidelity at a fraction of the compute cost.

2. Technical Approach

The Aerivio edge module ingests raw hydrophone-array streams, performs on-device feature extraction, and runs a quantized CNN classifier whose outputs feed directly into the USV autonomy bus. The end-to-end architecture is shown in Figure 1; all inference occurs on-vessel.

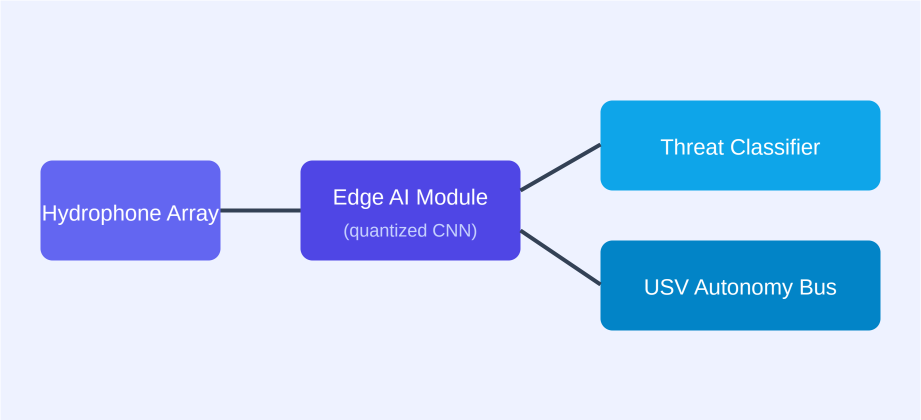

Figure 1. Aerivio edge acoustic-sensing architecture. All inference occurs on-vessel at 2 watts.

The approach rests on three co-designed innovations, detailed below.

2.1 Quantization-Aware Training

Rather than train in full precision and quantize afterward — which strands accuracy — we train with simulated 4-bit weight and 8-bit activation precision in the loop. Gradients flow through a straight-through estimator so the network learns weights that are robust to quantization noise. Preliminary experiments hold accuracy within 1.4 points of the full-precision baseline at 4-bit precision, versus a 9-point drop for post-training quantization.

2.2 Saliency-Guided Structured Pruning

Generic pruning removes parameters by magnitude and indiscriminately damages the spectral bands that separate confusable classes (for example, a propeller cavitation signature from a biologic click train). Our saliency metric scores each channel by its contribution to inter-class separation in the confusion-prone subspace, and prunes only channels below a learned threshold. This removes roughly 70% of parameters while preserving the discriminative bands — the core of Aerivio's defensible IP (patent pending).

2.3 Deterministic Real-Time Scheduler

Accuracy is necessary but not sufficient; the classifier must deliver a decision within a hard deadline at a fixed power budget. We implement a static, deadline-driven scheduler that pins feature extraction and inference to reserved cores, eliminating jitter from dynamic frequency scaling. This guarantees sub-100ms end-to-end latency at a measured 2.0-watt draw.

2.4 Preliminary Feasibility Evidence

Bench experiments on a representative subset of the DARPA hydrophone corpus already indicate feasibility. Table 1 summarizes the baseline-versus-target comparison that Phase I will formalize and extend to the full corpus and the target hardware.

Metric	Full-precision baseline	Aerivio 4-bit target
Classification accuracy	93.6%	**92%+**
Parameters	4.1 M	1.2 M
End-to-end latency	310 ms	**<100 ms**
Power draw	11 W	**2.0 W**

Table 1. Preliminary baseline vs. Phase I target (representative corpus subset).

3. Phase I Objectives

The technical objectives of Phase I are to establish feasibility of the edge classifier against Navy-relevant criteria:

1. Objective 1 — Curate and augment the hydrophone training corpus with documented provenance and class balance (supports Task 1).
2. Objective 2 — Train and quantize the baseline classifier to 4-bit precision, holding accuracy above 92% (supports Task 2).
3. Objective 3 — Port the runtime to the target 2-watt edge module and benchmark end-to-end latency (supports Task 3).
4. Objective 4 — Validate accuracy, latency, and power against Navy-relevant threat scenarios and document results (supports Task 4).

4. Phase I Work Plan

The six-month effort is organized into four sequential-but-overlapping tasks. Table 2 gives the schedule; the narrative that follows details scope, methods, and deliverables per task.

Task	Description	Months	Deliverable
1	Corpus curation & augmentation	1–2	Labeled dataset + data card
2	Quantization-aware training	2–4	Trained 4-bit model + accuracy report
3	Edge port & latency benchmark	4–5	Runtime on target module + benchmark log
4	Validation & final report	5–6	Phase I feasibility report

Table 2. Phase I task schedule (6-month period of performance).

Task 1 — Corpus Curation and Augmentation (Months 1–2)

We will assemble a labeled training corpus from the DARPA hydrophone release, supplemented by augmentation that models realistic sea-state noise, Doppler shift, and multipath. Each class will be balanced and every record documented in a data card capturing provenance and known limitations. Deliverable: the labeled dataset and its data card.

Task 2 — Quantization-Aware Training (Months 2–4)

Using the curated corpus, we will train the baseline classifier with 4-bit quantization and saliency-guided pruning in the loop, iterating on the saliency threshold to maximize inter-class separation under the parameter budget. Deliverable: the trained 4-bit model and an accuracy report against a held-out test split.

Task 3 — Edge Port and Latency Benchmark (Months 4–5)

We will port the runtime to the target 2-watt module, integrate the deterministic scheduler, and benchmark end-to-end latency and power under sustained load. Deliverable: the runtime on the target module and a benchmark log demonstrating the latency and power envelope.

Task 4 — Validation and Final Report (Months 5–6)

We will validate accuracy, latency, and power against a set of Navy-relevant threat scenarios and document the results, limitations, and the Phase II plan. Deliverable: the Phase I feasibility report.

5. Evaluation Methodology

Feasibility will be judged against pre-registered quantitative gates, not qualitative impressions. Accuracy is measured as balanced multi-class F1 on a held-out split drawn from distinct recording sessions to prevent leakage. Latency is measured end-to-end on the target hardware at the 99th percentile under sustained load, not as a mean. Power is measured at the module rail with a calibrated meter. A pass requires $F1 \geq 0.92$, p99 latency < 100 ms, and sustained draw ≤ 2.0 W simultaneously.

6. Risk Assessment and Mitigation

Risk	Likelihood	Mitigation
4-bit accuracy shortfall	Medium	Mixed-precision fallback on the most confusable layers
Latency exceeds budget on target	Low	Reserve cores + static schedule; profile early in Task 3
Corpus class imbalance	Medium	Targeted augmentation + class-balanced sampling in Task 1
Hardware availability slip	Low	Two development kits procured up front (see Cost Volume)

Table 3. Principal Phase I risks and mitigations.

7. Related Work and Innovation

Prior edge-audio efforts target keyword spotting at low fidelity and small class counts; none address multi-class maritime threat discrimination under a hard power ceiling. Post-training quantization methods are simple but strand accuracy on confusable classes. Magnitude pruning is standard but blind to inter-class saliency. Aerivio's contribution is the coupling of quantization-aware training with a saliency-guided pruning metric specific to acoustic threat discrimination — a combination we have validated in preliminary bench tests and are pursuing as protected IP.

8. Team and Facilities

The effort is led by Dr. Mara Ellison (Principal Investigator, 0.30 FTE), with Senior Systems Engineer Tomas Reyes and ML Engineer Priya Nandakumar. Full biographies appear in the Key Personnel supporting document. Aerivio maintains a dedicated acoustics bench and anechoic enclosure; facilities and equipment are described in the Facilities supporting document. Data collection leverages an existing marine-research-station partnership at no additional Phase I cost.

9. Commercialization Summary

The Phase I classifier is the wedge for a broad commercialization plan detailed in the Commercialization volume: Navy USV programs of record in Phase III, then allied navies and commercial maritime security (port monitoring, offshore-energy protection). The serviceable market is projected to grow from roughly $8M/yr in 2026 to over $60M/yr by 2030.

References

- [1] DARPA Hydrophone Corpus, Distribution A, 2024.
- [2] Ellison, M. et al., "Saliency-Guided Quantization for Acoustic Edge Inference," 2025 (in preparation).
- [3] Reyes, T., "Deterministic Scheduling for Sub-Watt Inference," Embedded ML Workshop, 2024.
- [4] Nandakumar, P., "Sea-State Augmentation for Hydrophone Classifiers," 2025 (in preparation).
- [5] U.S. Navy, "Unmanned Campaign Framework," 2023.
- [6] Jacob, B. et al., "Quantization and Training of Neural Networks for Efficient Integer-Arithmetic Inference," 2018.